

Introduction to Cloud Security

Basics of Cloud Security

Cloud security refers to the policies, technologies, and controls used to protect data, applications, and infrastructure in cloud environments. Major cloud providers like AWS, Azure, and Google Cloud (GCP) offer security tools and follow industry best practices to protect data from threats.

Shared Responsibility Model

In cloud computing, security is a shared responsibility between the cloud provider and the customer.

- Cloud Provider: Secures the infrastructure (hardware, software, networking).
- Customer: Secures data, applications, user access, and configurations.

Cloud Security Threats

Cloud computing, while powerful and flexible, brings along various security risks. Some general cloud security threats include:

- Misconfigurations
- Unauthorized access or weak access controls
- Data breaches or leaks
- Insider threats
- Insecure APIs
- Denial of Service (DoS) attacks
- Compliance violations
- Lack of visibility and monitoring

Common Cloud Security Threats (with Real-World Examples)

1. Misconfigurations

What it is: Mistakes in cloud settings (like open S3 buckets or weak IAM policies) that expose data to unauthorized access.

Real-world Example:

In 2017, Accenture exposed sensitive customer data due to misconfigured AWS S3 buckets. These buckets were left publicly accessible without authentication, exposing data such as API keys and customer details.

Impact: Data leaks, reputation damage, and possible regulatory fines.

2. Data Breaches / Leaks

What it is: Unauthorized access to sensitive data stored in the cloud due to poor encryption, access control, or insider threats.

Real-world Example:

In 2019, Capital One suffered a breach when a former AWS employee exploited a firewall misconfiguration to access over 100 million customer records.

Impact: Financial loss, customer trust issues, and lawsuits.

3. Insider Threats

What it is: Employees or third-party vendors misusing access to cloud resources, either maliciously or by accident.

Real-world Example:

In 2020, an AWS engineer was found to have intentionally accessed servers to steal customer data. Though detected early, this highlighted how dangerous insider threats can be.

Impact: Sensitive data loss, difficult to detect, potential long-term damage.